Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
AI-501
Fall 2025
Mid Exam – Part II

Weightage: 60%

Name: _____

Campus ID: _____

Note:
There are ten questions worth 30 points. You are required to attempt all questions. Use the back of pages for additional space to write your answers.

Problem 1. (**3 pts**) Consider a medical image from an MRI scan represented as a matrix X of size 2048×1536 (rows $\times$ columns). Assume that the matrix X has only 80 non-zero singular values. Using a **low-rank approximation** with rank $k = 80$, determine the **storage savings** achieved in terms of the **number of pixels** (or matrix entries) that need to be stored.

(a) What is the total number of entries in the original matrix X?

(b) How many entries need to be stored for the low-rank approximation of rank $k = 80$?

(c) Calculate the percentage reduction in storage enabled by this low-rank approximation.

Solution:

1. **Original Entries:** $2048 \times 1536 = 3,145,728$
2. **Low-Rank Storage:** The storage for a rank-k approximation using the SVD is given by storing the U_k, S_k, and V_k^T matrices.

$$\begin{aligned} U_k &\text{ is } 2048 \times 80 \\ S_k &\text{ is } 80 \times 1 \text{ (only the diagonal values)} \\ V_k^T &\text{ is } 80 \times 1536 \\ \text{Total} &= (2048 \times 80) + 80 + (80 \times 1536) \\ &= 163,840 + 80 + 122,880 = 286,800 \end{aligned}$$

3. **Percentage Reduction:**

$$\begin{aligned} \text{Reduction} &= \frac{\text{Original} - \text{Compressed}}{\text{Original}} \times 100\% \\ &= \frac{3,145,728 - 286,800}{3,145,728} \times 100\% \approx 90.88\% \end{aligned}$$

Problem 2. (**2 pts**) Suppose A is a 6×12 matrix, B is a 12×8 matrix, C is a 8×8 matrix, and D is a 6×6 matrix. Determine whether the following expression makes sense. If the expression makes sense, give its dimensions.

$A^T DA + BC^2 B^T$

Solution:

Given dimensions:

- A: 6×12
- B: 12×8
- C: 8×8
- D: 6×6

First term: $A^T DA$

$$A^T : 12 \times 6$$
$$D : 6 \times 6$$
$$A : 6 \times 12$$
$$A^T D : (12 \times 6)(6 \times 6) = 12 \times 6$$
$$(A^T D)A : (12 \times 6)(6 \times 12) = 12 \times 12$$

Second term: $BC^2 B^T$

$$B : 12 \times 8$$
$$C : 8 \times 8$$
$$C^2 : (8 \times 8)(8 \times 8) = 8 \times 8$$
$$B^T : 8 \times 12$$
$$BC^2 : (12 \times 8)(8 \times 8) = 12 \times 8$$
$$(BC^2)B^T : (12 \times 8)(8 \times 12) = 12 \times 12$$

Both terms are 12×12, so the addition is valid.
Dimensions: 12×12

Problem 3. (**3 pts**)

Let $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Apply Gram–Schmidt to construct an orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2\}$ spanning the same subspace.

Solution: *Step 1 (First vector).* $\mathbf{q}_1 = \dfrac{\mathbf{u}_1}{\|\mathbf{u}_1\|_2} = \dfrac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

Step 2 (Orthogonalize $\mathbf{u}_2$). $\mathbf{w}_2 = \mathbf{u}_2 - (\mathbf{q}_1^\top \mathbf{u}_2)\mathbf{q}_1$. Compute $\mathbf{q}_1^\top \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1 \cdot 1 + 1 \cdot 0 + 0 \cdot 1) = \frac{1}{\sqrt{2}}$. Thus

$$\mathbf{w}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}.$$

Step 3 (Normalize). $\|\mathbf{w}_2\|_2 = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$, so $\mathbf{q}_2 = \dfrac{1}{\sqrt{3/2}} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \sqrt{\frac{2}{3}} \end{bmatrix}$.

Problem 4. (**2 pts**)

A company's inventory is represented by a 4-vector $\mathbf{q}$, where each component q_i represents the quantity of product i in stock.

The cost (in dollars) per unit of each product is given by the vector $\mathbf{c} = \begin{bmatrix} 10 \\ 25 \\ 15 \\ 30 \end{bmatrix}$. The weight (in kg) per unit of each product is given by the vector $\mathbf{w} = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 5 \end{bmatrix}$.

Using vector notation and operations:

(a) Find an expression for the total value of the entire inventory.

(b) Find an expression for the total weight of the entire inventory.

(c) If the inventory vector is $\mathbf{q} = \begin{bmatrix} 20 \\ 10 \\ 15 \\ 5 \end{bmatrix}$, compute the total value and total weight.

Solution:

(a) The total value of the inventory is given by the dot product

$$\boxed{V = \mathbf{c}^\mathsf{T}\mathbf{q}}.$$

(b) The total weight of the inventory is

$$\boxed{W = \mathbf{w}^\mathsf{T}\mathbf{q}}.$$

(c)

$$V = 825 \text{ kg}, \quad W = 105 \text{ kg}.$$

Problem 5. (**3 pts**)

You are given a training dataset for predicting the category of a text document based on the counts of two words. The feature vectors (Word1, Word2) and labels (Category A or B) are:

- $(1, 2)$ and $(2, 1)$ for category A
- $(3, 4)$, $(4, 3)$, $(5, 5)$ for category B

A new test point has the feature vector $(2, 3)$. Using the k-Nearest Neighbor (kNN) algorithm with $k = 3$ and Euclidean distance, determine the predicted category for the test point. Show your work (list the distances and the 3 nearest neighbors). In case of two neighbours at the same distance, use a rule to pick the first neighbour in the ordering.

Let the test point be $x^* = (2, 3)$. Euclidean distances to the training points:

$$\begin{aligned}
d((1,2), A) &= \sqrt{(2-1)^2 + (3-2)^2} = \sqrt{2} \approx 1.414, \\
d((2,1), A) &= \sqrt{(2-2)^2 + (3-1)^2} = 2, \\
d((3,4), B) &= \sqrt{(2-3)^2 + (3-4)^2} = \sqrt{2} \approx 1.414, \\
d((4,3), B) &= \sqrt{(2-4)^2 + (3-3)^2} = 2, \\
d((5,5), B) &= \sqrt{(2-5)^2 + (3-5)^2} = \sqrt{13} \approx 3.606.
\end{aligned}$$

Tie-breaking rule: when multiple points are at the same distance, choose the one that appears *earlier* in the given list: $(1,2)A \prec (2,1)A \prec (3,4)B \prec (4,3)B \prec (5,5)B$.

The sorted distances (with ties broken by the rule) give the three neighbors: $(1,2), A$ ($\sqrt{2}$), $(3,4), B$ ($\sqrt{2}$), and then between distance 2 ties we pick $(2,1), A$ before $(4,3), B$. Thus the 3 neighbors are $\{A, B, A\} \Rightarrow$ majority A.

$$\boxed{\text{Predicted category for } k = 3 : \ A}$$

Problem 6. (**2 pts**)

Suppose A is a 5×7 matrix with rank 4.

(a) What is the dimension of the column space $\mathcal{C}(A)$?

(b) What is the dimension of the null space $\mathcal{N}(A)$? Justify your answer using the Rank–Nullity Theorem.

(c) Is the system $A\mathbf{x} = \mathbf{b}$ guaranteed to have a solution for every $\mathbf{b} \in \mathbb{R}^5$? Explain why or why not.

Given $A \in \mathbb{R}^{5\times 7}$ with $\text{rank}(A) = 4$.

(a) The dimension of the column space equals the rank:

$$\dim \mathcal{C}(A) = \text{rank}(A) = \boxed{4}.$$

(b) By the Rank–Nullity Theorem for a 5×7 matrix (i.e., $n = 7$ columns),

$$\underbrace{\text{rank}(A)}_{4} + \underbrace{\text{nullity}(A)}_{\dim \mathcal{N}(A)} = n = 7 \quad \Rightarrow \quad \dim \mathcal{N}(A) = 7 - 4 = \boxed{3}.$$

(c) The system $A\mathbf{x} = \mathbf{b}$ has a solution for a given $\mathbf{b} \in \mathbb{R}^5$ iff $\mathbf{b} \in \mathcal{C}(A)$. Since $\mathcal{C}(A)$ is a *proper* subspace of $\mathbb{R}^5$ with dimension $4 < 5$, it does *not* equal $\mathbb{R}^5$. Therefore, $A\mathbf{x} = \mathbf{b}$ is *not* guaranteed to have a solution for every $\mathbf{b} \in \mathbb{R}^5$; only those $\mathbf{b}$ lying in the column space of A are attainable. Equivalently, for the system to be solvable for every $\mathbf{b}$, we would need $\text{rank}(A) = 5$, which is not the case here.

$$\boxed{\text{No; only } \mathbf{b} \in \mathcal{C}(A) \text{ yield solutions.}}$$

Problem 7. (**5 pts**)

(a) Write down the loss function for Ridge regression, and derive the closed-form solution for the ridge regression estimator.

(b) Let

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad Y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \lambda = 1.$$

Compute the ridge regression estimate.

(c) How does the value of λ influence the magnitude of the coefficients $\mathbf{w}_{\text{ridge}}$? What happens when $\lambda \to \infty$?

(a) **Loss and closed form.** Ridge regression solves

$$\mathcal{L}_{\text{ridge}}(\mathbf{w}) = \frac{1}{2} \|Y - X\mathbf{w}\|_2^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2.$$

Set the gradient to zero:

$$-X^T(Y - X\mathbf{w}) + \lambda\mathbf{w} = 0 \implies (X^TX + \lambda I)\mathbf{w} = X^TY.$$

Thus

$$\boxed{\mathbf{w}_{\text{ridge}} = (X^TX + \lambda I)^{-1}X^TY}.$$

(b) **Numerical estimate for** $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $Y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\lambda = 1$.

$$X^TX = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}, \quad X^TX + \lambda I = \begin{bmatrix} 3 & 3 \\ 3 & 6 \end{bmatrix}, \quad (X^TX + \lambda I)^{-1} = \frac{1}{9}\begin{bmatrix} 6 & -3 \\ -3 & 3 \end{bmatrix}.$$

Also $X^TY = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. Hence

$$\mathbf{w}_{\text{ridge}} = (X^TX + \lambda I)^{-1}X^TY = \frac{1}{9}\begin{bmatrix} 6 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \boxed{\begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}}.$$

(c) **Effect of λ.** As λ increases, the penalty $\frac{\lambda}{2}\|\mathbf{w}\|_2^2$ forces coefficients toward zero (more bias, less variance). In the limit,

$$\boxed{\lambda \to \infty \;\Rightarrow\; \mathbf{w}_{\text{ridge}} \to \mathbf{0}}.$$

Problem 8. (**5 pts**) Consider the matrix $C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

(a) Without performing detailed calculations, explain why the vectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ are likely to be eigenvectors of C.

(b) Based on your answer in (a), determine **orthogonal** matrix Q and eigenvalues Λ such that $C = Q\Lambda Q^{-1}$.

(c) **Using the EVD**, compute C^2.

Let $C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

(a) **Why $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ are eigenvectors.** The matrix C is symmetric with equal diagonal entries and equal off–diagonal entries. Such matrices act as:

$$C\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x+y \\ x+2y \end{bmatrix} = \underbrace{(x+y)}_{\text{sum}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \underbrace{(x-y)}_{\text{difference}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Hence the "sum" direction $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the "difference" direction $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ are invariant directions, i.e., eigenvectors.

(b) **EVD $C = Q\Lambda Q^{-1}$.** Compute the eigenvalues:

$$C\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \lambda_1 = 3, \qquad C\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1\begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \Rightarrow \quad \lambda_2 = 1.$$

Choose orthonormal eigenvectors

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Then

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

and since C is symmetric, $Q^{-1} = Q^T$ and $C = Q\Lambda Q^T$.

(c) **Compute C^2 via EVD.**

$$C^2 = Q\,\Lambda^2\,Q^T = Q\begin{bmatrix}9 & 0\\ 0 & 1\end{bmatrix}Q^T = \frac{1}{2}\left(\begin{bmatrix}9 & 9\\ 9 & 9\end{bmatrix} + \begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\right) = \begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix}.$$

$$\boxed{C^2 = \begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix}}.$$

Problem 9. (**3 pts**) The SVD provides a powerful geometric interpretation of a linear transformation.

(a) Describe the three-step geometric transformation (rotation/reflection, scaling, rotation/reflection) represented by the SVD of a matrix A.

(b) A 2×2 matrix C transforms the unit circle into an ellipse. How are the left singular vectors U, the right singular vectors V, and the singular values Σ related to the orientation and axes lengths of this ellipse?

Solution: For a matrix $A \in \mathbb{R}^{m\times n}$ with SVD $A = U\Sigma V^T$, the transformation $A\mathbf{x}$ can be interpreted as:

1. **Rotation/Reflection in $\mathbb{R}^n$:** Multiply by V^T — transforms the input into the coordinate system of the right singular vectors.
2. **Scaling in $\mathbb{R}^n$ (and projection/padding if $m \neq n$):** Multiply by Σ — stretches or compresses along the coordinate axes by the singular values σ_i.
3. **Rotation/Reflection in $\mathbb{R}^m$:** Multiply by U — aligns the output with the left singular vectors in $\mathbb{R}^m$.

Let $C = U\Sigma V^T$. Then:

- The **right singular vectors** (columns of V) give the directions of the principal axes of the ellipse in the input space.
- The **singular values** (diagonal entries of Σ) give the lengths of the semi-axes of the ellipse.
- The **left singular vectors** (columns of U) give the directions of the principal axes of the ellipse in the output space.

In particular, the unit circle $\|\mathbf{x}\| = 1$ is mapped to an ellipse whose semi-axes are $\sigma_1\mathbf{u}_1$ and $\sigma_2\mathbf{u}_2$.

Problem 10. (**2 pts**) Suppose the SVD of a matrix $B \in \mathbb{R}^{3\times 2}$ is given by:

$$U = \begin{bmatrix}1/\sqrt{2} & 1/\sqrt{2} & 0\\ 1/\sqrt{2} & -1/\sqrt{2} & 0\\ 0 & 0 & 1\end{bmatrix}, \quad \Sigma = \begin{bmatrix}5 & 0\\ 0 & 2\\ 0 & 0\end{bmatrix}, \quad V = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

(a) What are the eigenvalues of B^TB?

(b) Compute the best rank-1 approximation B_1 of the matrix B.

Given the SVD $B = U\Sigma V^\top$ with

$$U = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0\\ 0 & 0 & 1\end{bmatrix}, \quad \Sigma = \begin{bmatrix}5 & 0\\ 0 & 2\\ 0 & 0\end{bmatrix}, \quad V = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

(a) **Eigenvalues of $B^\top B$.**

$$B^\top B = V\Sigma^\top \Sigma V^\top = V \begin{bmatrix} 25 & 0 \\ 0 & 4 \end{bmatrix} V^\top.$$

Since $V = I_2$, the eigenvalues are simply the diagonal entries:

$$\boxed{\lambda_1 = 25,\ \lambda_2 = 4}.$$

(b) The best rank-1 approximation B_1 of the matrix B is given by:

$$B_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$$

where $\mathbf{u}_1$ is the first column of U and $\mathbf{v}_1$ is the first column of V.
From the given matrices:

$$\mathbf{u}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \sigma_1 = 5$$

First compute the outer product:

$$\mathbf{u}_1 \mathbf{v}_1^T = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}$$

Now multiply by $\sigma_1 = 5$:

$$B_1 = 5 \cdot \begin{bmatrix} 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 5/\sqrt{2} & 0 \\ 5/\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}$$

Alternatively, rationalizing the denominator:

$$B_1 = \begin{bmatrix} \frac{5\sqrt{2}}{2} & 0 \\ \frac{5\sqrt{2}}{2} & 0 \\ 0 & 0 \end{bmatrix}$$